

SESSIONAL EXAMINATION - I

EVEN SEMESTER, 2024-2025

Branch: CSE	Batch: 2023	Semester: IV
Course Title: Machine Learning System for Real World	Course Code	23CS006
ID. No. of Student	Max. Marks	40
2 3 1 1 9 8 1 0 0 7	Max Duration	90 Minutes

Instructions:

- Attempt questions in order.
- Students not to write anything on question paper.
- Calculator allowed.

Course Learning Outcomes:

CLO-1	Understand concepts associated with machine learning algorithms
CLO-2	Recognize the practical benefits of machine learning.
CLO-3	Analyse the concept of neural networks for learning linear and non-linear activation functions.
CLO-4	Understand the trade-offs between complexity and generalizability.
CLO-5	Design and implement various machine learning algorithms in a range of real-world applications.

SECTION - A

(All Questions are Compulsory, Each question carries 01 mark) (5x1=5)

- A bank wants to predict whether a customer will repay a loan on time based on their credit history and income. **CLO-1**
Identify the type of machine learning technique should be used?
 (i) Reinforcement Learning (ii) Generative AI
 (iii) Supervised Learning (iv) UnSupervised Learning
- function in scikit-learn is used to split data into training and testing sets. **CLO-1**
 (i) split_train_test() (ii) split_data()
 (iii) divide_data() (iv) train_test_split()
- Which of the following technique is used to handle overfitting in decision trees? **CLO-2**
 (i) Data augmentation (ii) Pruning
 (iii) Feature scaling (iv) Normalization
- Given a dataset with correlation coefficient $r = 0.8$, what is the coefficient of determination (R^2)? **CLO-1**
 (i) 0.4 (ii) 0.8
 (iii) 0.16 (iv) 0.64
- Which metric is used to evaluate the performance of a classification model? **CLO-1**
 (i) Mean Absolute Error (ii) Confusion matrix
 (iii) Gradient Descent (iv) Clustering Index

SECTION – B

(Attempt any 5 questions, each question carries 02 marks) (5x2=10)

6. How would `info()` and `describe()` methods in Pandas help in analyzing a dataset? CLO-2
7. A real estate agency uses a linear regression model to predict house prices based on various features like location, size, and age of the property. To evaluate how well the model explains the variation in house prices, the agency calculates the R-square value. What does the R-square value represent in this scenario? CLO-2
8. In a simple linear regression model, the Sum of Squared Errors (SSE) is 50, and the Sum of Squares Regression (SSR) is 200. The Total Sum of Squares (SST) is not directly provided. Calculate R^2 . CLO-1
9. A fraud detection system was tested to detect fraudulent transactions. The confusion matrix obtained from the model's predictions is as follows: CLO-2

	Predicted Fraud	Predicted F raud
Actual Fraud	80 TP	20 FN
Actual Not Fraud	15 FP	150 TN

Calculate the following performance metrics: accuracy, precision, F1-score, recall.

10. A detailed and thorough explanation, touching upon the theoretical foundations, practical applications, and nuances of both linear and non-linear regression models. CLO-3

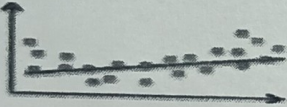
11. CLO-2
- 

Figure 1

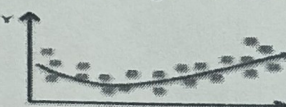


Figure 2

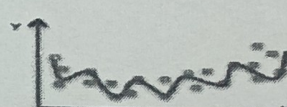


Figure 3

Understanding model fit is essential for predicting model accuracy. Based on the provided graph, how can you determine the type of fit the model is demonstrating, and mention the key characteristics of that fit?

SECTION – C

(Attempt any 3 questions, each question carries 5 marks) (3x5=15)

12. Differentiate between Lasso and Ridge regression in terms of feature selection, regularization type, output model, impact on prediction, computation. CLO-3
13. Assume, you are given a dataset `data.csv`. Write a Python code to: CLO-4
- Load the dataset.
 - Display the last 5 rows.
 - Print basic information about the dataset.
 - Check for missing values in the dataset.
 - Display basic statistics for numeric columns.

14. A data scientist at a medical research company has developed a model to predict whether a patient is at high risk of developing a disease. The model outputs a predicted probability (between 0 and 1) for each patient, indicating the likelihood of developing the disease. To evaluate the performance of your model in identifying high-risk patients using an ROC curve. The company has collected data for 10 patients and their corresponding predicted probabilities, as shown in the table below: **CLO-3**

TP- True positive

FP- False positive

TPR- True positive rate

FPR- False positive rate

ROC-Receiver Operating Characteristic

AUC- Area Under the Curve

Actual	Predict Probability	TP	FP	TPR	FPR
True	0.90				
True	0.80				
False	0.70				
True	0.60				
True	0.55				
False	0.54				
False	0.53				
False	0.51				
True	0.50				
False	0.40				

Construct a ROC curve to evaluate the performance of your model in distinguishing high-risk patients (True) from low-risk patients (False).

- Plot the ROC curve based on the calculated TPR and FPR values.
- Interpret the ROC curve to understand the model's performance.
- Also calculate the AUC.

15. A real estate company is building a model to estimate home prices using features like bedrooms, square footage, location, year built, and amenities, based on data from over 1500 properties. The company is concerned about overfitting due to many features, some of which may be irrelevant or highly correlated. Considering the circumstances, which regularization technique—Lasso or Ridge Regression—would be more appropriate for this scenario? Discuss the suitability of each technique with relevant examples. **CLO-4**

SECTION – D

(Attempt any one question, each question carries 10 marks, subparts (if any) carry equal weightage) (1x10=10)

16. Suppose a decision tree is being built for a binary classification problem to predict whether a customer will buy a product (Yes or No). The dataset contains a feature "Weather" with the following values: CLO-4

Weather	Buy a Product (Yes/No)
Sunny	Yes
Sunny	No
Rainy	Yes
Sunny	Yes
Rainy	No
Rainy	Yes
Sunny	No
Rainy	Yes

- a) Calculate the Entropy before and after split of the dataset.
b) Calculate the Information Gain for splitting on the "Weather" feature.
17. A data scientist is attempting to forecast product sales based on advertising expenditure. After developing the regression model, they obtain the following actual and predicted values for five data points. CLO-2

Advertising Expenditure (in Rs1000)	Actual Sales (in Rs1000)	Predicted Sales (in Rs1000)
5	15	14
10	25	24
15	35	34
20	45	43
25	55	52

Calculate the following:

- a) Mean Squared Error (MSE)
b) Root Mean Squared Error (RMSE)
c) Correlation Coefficient
d) R-squared (R^2)

SESSIONAL EXAMINATION - II

EVEN SEMESTER, 2024-2025

Branch : CSE		Batch : 2023	Semester : 4 th
Course Title	Machine Learning System for Real World	Course Code	23CS006
ID. No. of Student		Max. Marks	40
2311981007		Max Duration	90 Minutes

Instructions:

- Attempt questions in order.
- Students not to write anything on question paper.
- Calculator allowed.

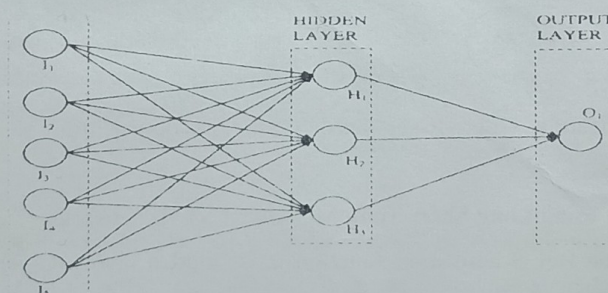
Course Learning Outcomes:

CLO-1	Understand concepts associated with machine learning algorithms
CLO-2	Recognize the practical benefits of machine learning.
CLO-3	Analyse the concept of neural networks for learning linear and non-linear activation functions.
CLO-4	Understand the trade-offs between complexity and generalizability.
CLO-5	Design and implement various machine learning algorithms in a range of real-world applications.

SECTION - A

(All Questions are Compulsory, Each question carries 01 mark) (5x1=5)

- Given a labeled binary classification data set with N data points and D features. Suppose that $N < D$. In training an support vector machine (SVM) on this data set, which of the following kernels is likely to be most appropriate? CLO-1
 - Linear kernel
 - Quadratic kernel
 - Higher-order polynomial kernel
 - Radial Basis Function kernel
- Consider the following image and select the correct option below: CLO-3



- Total number of layers in this model is 5
- Total number of neurons in the model is 9
- Total number of hidden layers is 3
- Total number of neurons in output layer is 3

3. Which of the following is not a typical component of an artificial neural network?
- (i) Neurons (ii) Convolutional layers
(iii) Activation functions (iv) Weights
4. In the context of Long Short-Term Memory (LSTM), mention the role of the cell state.
- (i) It helps propagate the gradients during back-propagation (ii) It stores information for long-term memory, helping the network remember important data over time.
(iii) It decides which output should be passed to the next time step (iv) It is used to initialize the weights of the network
5. Which of the following is a commonly used pooling method in Convolutional Neural Networks (CNNs)?
- (i) Mean pooling (ii) Max pooling
(iii) Min pooling (iv) Sum pooling

CLO-4

CLO-3

CLO-1

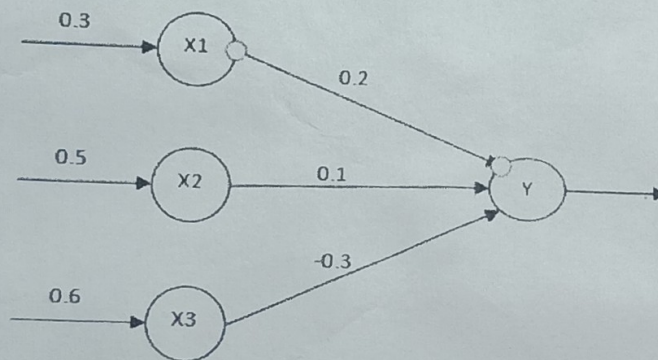
SECTION – B

(Attempt any 5 questions, each question carries 02 marks) (5x2=10)

6. A support vector machine (SVM) is trained to classify emails as "spam" or "not spam." After deploying the model, it performs poorly on new emails containing slang and informal language. These slangs and other informal language keywords were absent in training dataset. Identify the most likely cause for poor performance.
7. Calculate the output of the given neural network using sigmoid function. Consider bias as '0'.

CLO-1

CLO-3



8. Describe two real-world applications where RNNs (or their variants) outperform CNNs or feed-forward networks. Justify your answer.
9. Write the main advantages of using Long Short-Term Memory (LSTM) networks.
10. A neural network is using the ReLU activation function in its hidden layers. Given the following input values [4, -3, 0, 6, -2]. Calculate the output after applying the ReLU function.
11. In a Convolutional Neural Network (CNN), mention the primary purpose of the convolutional layer and how does it differ from a fully connected layer?

CLO-1

CLO-2

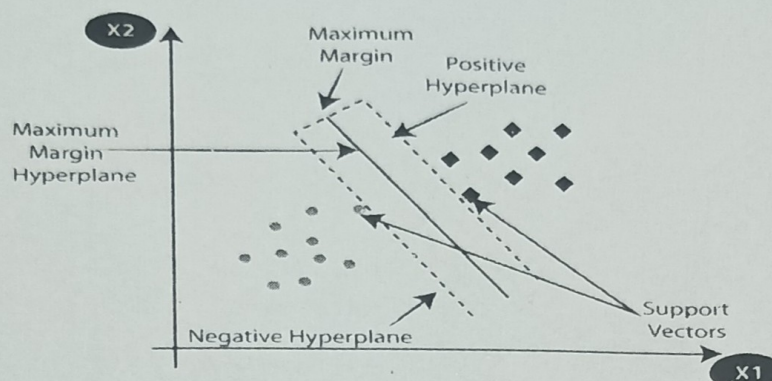
CLO-3

CLO-2

SECTION – C

(Attempt any 3 questions, each question carries 5 marks) (3x5=15)

12. Differentiate between convolutional neural network and recurrent neural network in terms of memory, architecture, parallelization, use cases, type of data, training difficulty. **CLO-1**
13. Explain all the key steps of the support vector machine (SVM) algorithm based on the diagram provided below. **CLO-3**



14. A delivery company wants to predict whether a new delivery location will be classified as "Fast Delivery (F)" or "Slow Delivery (S)" based on past data. The classification depends on the location's distance from the warehouse and traffic density in that area. The company has collected data from previous deliveries: **CLO-3**

Distance from Warehouse (km)	Traffic Density (cars/hour)	Delivery Speed (Class)
2	30	F
5	60	S
8	80	S
4	80	F
7	75	S
3	35	?

Now, a new delivery location (3 km, 35 cars/hour) needs to be classified using KNN with $K = 3$ and Euclidean distance. Tasks:

- Explain Euclidean distance formula.
 - Calculate the Euclidean distances between the new location (3, 35) and all other locations.
 - Identify the 3 nearest neighbours based on distance.
 - Predict whether the new location falls into F or S based on majority voting.
15. Suppose you are building a decision tree for a binary classification problem where you want to predict whether a customer will buy a product (Yes or No). With the following available data samples, calculate the entropy and gini index. **CLO-4**

Age Group	Buy a Product (Yes/No)
Young	Yes
Young	No
Middle-aged	Yes
Old	No
Young	Yes
Middle-aged	No
Old	Yes
Old	Yes

SECTION - D

(Attempt any one question, each question carries 10 marks, subparts (if any) carry equal weightage) (1x10=10)

CLO-4

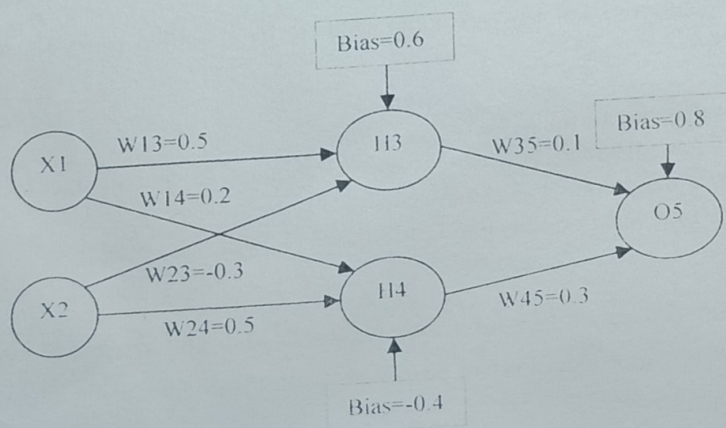
16. Given an input matrix X and a filter K , perform the following operations:
- Apply a convolution operation with a stride of 1 to obtain the feature map.
 - Apply the ReLU activation function to the feature map.
 - Perform max pooling with a 2×2 filter and stride 2 to obtain the final pooled feature map.
- Provide the resulting matrices at each step.

$$X = \begin{bmatrix} 2 & 4 & 6 & 8 & 10 \\ 1 & 3 & 5 & 7 & 9 \\ 0 & 2 & 4 & 6 & 8 \\ 1 & 5 & 9 & 13 & 17 \\ 3 & 4 & 9 & 12 & 15 \end{bmatrix}$$

$$K = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

17. Consider a multilayer feed-forward neural network given below. Let the learning rate be 0.5. Assume initial values of weights and biases as given in the diagram below. Train the network for the training tuple (0, 1, 1) where last number is target output. Calculate updated weights by using back-propagation algorithm. Assume that sigmoid activation function is used in the network. Perform one forward and backward pass.

CLO-4



SESSIONAL EXAMINATION - III

EVEN SEMESTER, 2024-2025

Branch : CSE	Batch : 2023	Semester : 4 th
Course Title : Machine Learning System for Real World	Course Code	23CS006
ID. No. of Student	Max. Marks	40
2 3 1 1 9 8 1 0 0 7	Max Duration	90 Minutes

Instructions:

- Attempt questions in order.
- Students not to write anything on question paper.
- Calculator allowed.

Course Learning Outcomes:

CLO-1	Understand concepts associated with machine learning algorithms
CLO-2	Recognize the practical benefits of machine learning.
CLO-3	Analyse the concept of neural networks for learning linear and non-linear activation functions.
CLO-4	Understand the trade-offs between complexity and generalizability.
CLO-5	Design and implement various machine learning algorithms in a range of real-world applications.

SECTION – A*(All Questions are Compulsory, Each question carries 01 mark) (5x1=5)*

- Which of the following techniques is most commonly used to minimize the loss function in neural networks? **CLO-1**
 - Cross-validation
 - Grid search
 - Gradient descent
 - Principal Component Analysis (PCA)
- Which machine learning algorithm predicts a patient's disease by identifying the 5 most similar historical cases and selecting the most common diagnosis among them? **CLO-1**
 - Naive Bayes
 - k-Nearest Neighbors
 - Random Forest
 - Support Vector Machine
- In a Long Short-Term Memory (LSTM), which component helps to control the flow of new information into the memory cell? **CLO-4**
 - Input gate
 - Forget gate
 - Output gate
 - Hidden state
- Given a dataset with correlation coefficient $r = 0.8$, what is the coefficient of determination (R^2)? **CLO-1**
 - 0.4
 - 0.8
 - 0.16
 - 0.64

5. Which framework is widely used for building and training deep learning models?
- (i) NumPy (ii) Keras
(iii) OpenCV (iv) SciPy

SECTION – B

(Attempt any 5 questions, each question carries 02 marks) (5x2=10)

6. A binary classifier gives the following results: TP=30, TN=50, FP=10, FN=10. Calculate the accuracy score. CLO-1
7. How does deep learning differ from traditional machine learning techniques? Justify this statement in terms of feature extraction. CLO-3
8. A student has a dataset containing the scores of 10 students in an exam, stored in a CSV file. The student wants to create a simple line chart to visualize the scores over the course of the semester. Which libraries will the student use to read and plot the dataset, and which basic functions from those libraries should be used to accomplish this? CLO-2
9. Calculate the Euclidean distance between two feature vectors (2, 3) and (4, 5). CLO-1
10. In a convolution neural network (CNN), state out the role of the pooling layer, and how does it contribute to the network's performance? CLO-2
11. Explain cross-validation and highlight the key differences between k-fold and stratified k-fold cross-validation. CLO-3

SECTION – C

(Attempt any 3 questions, each question carries 5 marks) (3x5=15)

12. A regression model is developed to forecast the amount of time a customer spends on a website based on various features. After testing the model following actual and predicted values were obtained for a sample of 6 customers. Calculate Mean Squared Error (MSE) and Root Mean Squared Error (RMSE). CLO-3

Customer	Actual (y)	Predicted (y _i)
1	30	32
2	45	50
3	60	55
4	80	75
5	100	95
6	120	118

13. Compare and contrast deep learning and machine learning technique. Describe at least two real-world applications. Mention commonly used frameworks and hardware for implementation. CLO-2
14. Assume, you are given a dataset named as "data.csv". Write a Python code to: CLO-4
- Load the dataset.
 - Display the first 5 rows.
 - Extract basic information about the dataset.
 - Check for missing values in the dataset.
 - Display basic statistics for numeric columns.

15. A real estate agent wants to predict the price of a house based on size (in square feet) and number of bedrooms. Historical data from previous sales is given below. The agent will use k-nearest neighbors (k-NN) regression to predict the price of a new house. Predict the price of house number 7, using $k=3$. **CLO-3**

House No.	Bedroom	House Size (Sq.ft)	Price (x1000)
1	3	1500	210
2	3	1700	220
3	4	1800	230
4	3	1900	240
5	4	2100	250
6	4	2300	260
7	4	2000	?

SECTION - D

(Attempt any one question, each question carries 10 marks, subparts (if any) carry equal weightage) (1x10=10)

16. A retail company wants to segment its customers based on their annual spending on groceries and annual spending on clothing (in multiple of Rs.1000). The company collected data of 6 customers provided in table below. **CLO-5**

Customer	Grocery Spending (x)	Clothing Spending (y)
A	2	10
B	3	12
C	8	3
D	9	2
E	4	11
F	10	2

Use k-means clustering to divide the customers into 2 clusters ($k=2$). Perform 1 iteration of the K-means algorithm starting with: Initial centroids: Cluster 1: (2, 10), Cluster 2: (10, 2). After making clusters compute new centroids.

17. A data analyst is working on a very small dataset that consists of only two observations with two features each. The data is represented as follows: **CLO-2**

$$X = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

The analyst wants to reduce the dimensionality of the data from 2D to 1D using principal component analysis (PCA). Perform PCA on the dataset step-by-step and find the transformed 1D data.

END TERM EXAMINATIONS
EVEN SEM, AY: 2024-2025

School/Branch	SOET	Batch: 2023, Semester: II	
Course Code	23CS006	Course Title	Machine Learning System for Real World
ID. No. of Student		Max. Marks	60
2311981007		Max Duration	3 Hours

Instructions:

- Attempt questions in order.
- Students not to write anything on question paper.
- Use of calculator is allowed.

Course Learning Outcomes:

CLO-1	Understand concepts associated with machine learning algorithms.
CLO-2	Recognize the practical benefits and research applications of machine learning.
CLO-3	Analyse the concept of neural networks for learning linear and non-linear activation function.
CLO-4	Apply analytical skills to process and interpret real time data effectively.
CLO-5	Design and implement machine learning algorithms in real world applications with a focus on sustainability.

SECTION – A

(Questions 1 to 5 has four choices, out of which only one is correct, no negative marking for incorrect answer, each question carries 01 mark) (5x1=5)

- If a CNN layer has 16 filters of size 5×5 and input depth=3 (RGB), how many trainable parameters does this layer have? **CLO-1**
 - 16
 - 1200
 - 1216
 - 2400
- A bank wants to predict whether a customer will repay a loan on time based on his/her credit history and income. Identify the type of machine learning technique to be used? **CLO-2**
 - Generative AI
 - Reinforcement Learning
 - UnSupervised Learning
 - Supervised Learning
- Which metric is used to evaluate the performance of a classification model? **CLO-3**
 - Mean Absolute Error
 - Confusion matrix
 - Gradient Descent
 - Clustering Index
- Given a labeled binary classification data set with N data points and D features. Suppose that $N < D$. In training a support vector machine (SVM) on this data set, which of the following kernels is likely to be most appropriate? **CLO-3**
 - Linear kernel
 - Quadratic kernel
 - Higher-order polynomial kernel
 - Radial Basis Function kernel
- Given a dataset with correlation coefficient $r = 0.8$, what is the coefficient of determination (R^2)? **CLO-2**
 - 0.4
 - 0.8
 - 0.16
 - 0.64

SECTION – B

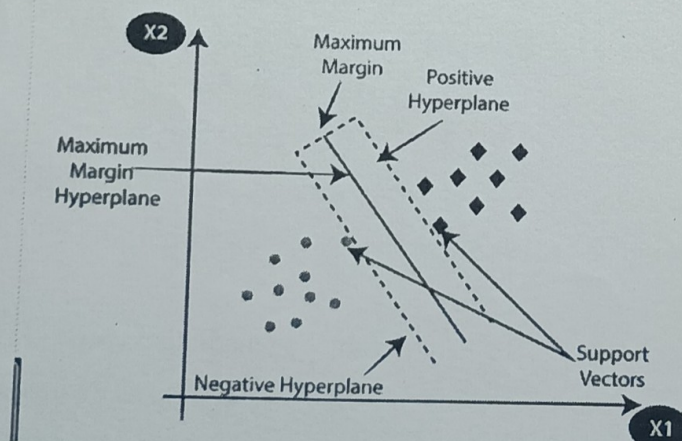
(Attempt any 5 questions, each question carries 02 marks) (5x2=10)

6. A binary classifier gives the following results: TP=30, TN=50, FP=10, FN=10. Calculate the accuracy score. CLO-1
7. Explain the types of artificial neural network with suitable examples. CLO-2
8. A neural network is using the ReLU activation function in its hidden layers. Given the following input values [4, -3, 0, 6, -2]. Calculate the output after applying the ReLU function. CLO-2
9. Explain the purpose of cross-validation and how it helps to prevent overfitting during model tuning. CLO-3
10. For a point at coordinates (3, 4) and its nearest neighbor at (5, 6), calculate the Euclidean distance for a point. CLO-2
11. Write a Python code to split a dataset into training (80%) and testing (20%) sets using scikit-learn. CLO-4

SECTION – C

(Attempt any 5 question, each question carries 05 marks, subparts (if any) carry equal weightage) (5x5=25)

12. Assume, you are given a dataset named as "data.csv". Write a Python code to:
a) Load the dataset.
b) Display the first 5 rows.
c) Extract basic information about the dataset.
d) Check for missing values in the dataset.
e) Display basic statistics for numeric columns. CLO-4
13. Differentiate between LASSO and Ridge regression in terms of feature selection, regularization type, output model, impact on prediction, computation. CLO-3
14. Explain all key features of the support vector machine (SVM) algorithm mentioned in given diagram. CLO-4



15. A model was tested on a binary classification task where the goal is to detect whether a patient has a particular disease. The confusion matrix obtained from the model's predictions is as follows: CLO-1

	Predicted Positive	Predicted Negative
Actual Positive	50	10
Actual Negative	5	100

Discuss all the metrics and also calculate the performance metrics: precision, recall, F1-score, accuracy, ROC.

16. Describe the architecture and working principle of Recurrent Neural Networks (RNNs) and mention two real-world applications. **CLO-1**
17. Suppose you are building a decision tree for a binary classification problem where you want to predict whether a customer will buy a product (Yes or No). The dataset contains a feature Age Group (Young, Middle-aged, Old) with the following values: **CLO-2**

Age Group	Buy a Product (Yes/No)
Young	Yes
Young	No
Middle-aged	Yes
Old	No
Young	Yes
Middle-aged	No
Old	Yes
Old	Yes

- Calculate the Entropy of the dataset (i.e., the entropy before the split).
- Calculate the Entropy after splitting the dataset containing "Age Group" feature.
- Calculate the Information Gain for the split on the "Age Group" feature.

SECTION – D

(Attempt any 2 question, each question carries 10 marks, subparts (if any) carry equal weightage) (2x10=20)

18. A data analyst is working on a very small dataset that consists of only two observations with two features each. The data is represented as follows: **CLO-**

$$X = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

The analyst wants to reduce the dimensionality of the data from 2D to 1D using principal component analysis (PCA). Perform step- by- step procedure to perform PCA and also find the

transformed 1D data.

19. A fitness app wants to group its users based on their average walking steps and average hours of sleep per day. The goal is to identify clusters for health-based recommendations. The app collected data from 6 users is provided in table below. **CLO-5**

User	Average Steps (in 1000s)	Average Sleep (hours)
U1	3	8
U2	4	7.5
U3	10	5
U4	11	4.5
U5	4.5	8
U6	12	5

Using k-means clustering ($k = 2$), perform 1 iteration of the algorithm using the following initial centroids: cluster 1: (3, 8), cluster 2: (11, 5). Compute new Centroid.

20. Given an input matrix "M" and a kernel "F", follow these steps: **CLO-4**
- Apply a convolution operation using a stride of 1 to generate the feature map matrix.
 - Apply the ReLU activation function on feature map to the obtained new feature map.
 - Use 2×2 max pooling and stride 2 to get the final pooled feature map.

Show the resultant matrix after each step.

$$M = \begin{bmatrix} 2 & 4 & 6 & 8 & 10 \\ 1 & 3 & 5 & 7 & 9 \\ 0 & 2 & 4 & 6 & 8 \\ 1 & 5 & 9 & 13 & 17 \\ 3 & 4 & 9 & 12 & 15 \end{bmatrix}$$

$$F = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$